

Class:

Name:

Time: 60 minutes

1. What is the directive that includes main() and scanf() functions?

A.stdio.h

B.math.h

C.iostream.h

D.system.in

2. What does the following declaration signify?

```
int *ptr[30];
```

A.ptr is a pointer to an array of 30 integer pointers.

B.ptr is an array of 30 pointers to integers.

C.ptr is an array of 30 integer pointers.

D.ptr is an array 30 pointers.

3. The keyword used to transfer control from a function back to the calling function is

A.switch

B.goto

C.go back

D.return

4. What will be the output of the program?

```

#include<stdio.h>
int main()
{
    int i = 1;
    switch(i)
    {
        printf("Hello\n");
        case 1:
            printf("Hi\n");
            break;
        case 2:
            printf("\nBye\n");
            break;
    }
    return 0;
}

```

A.

Hello

Hi

B.

Hello

Bye

C.Hi

D.Bye

5. What will be the output of the program?

```

#include<stdio.h>

int main()
{
    const char *s = "";
    char str[] = "Hello";
    s = str;
    while(*s)
        printf("%c", *s++);

    return 0;
}

```

A.Error

B.H

C.Hello

D.Hel

6. What will be the output of the program?

```
#include<stdio.h>
#include<string.h>

int main()
{
    printf("%c\n", "abcdefgh"[4]);
    return 0;
}
```

A.Error

B.d

C.e

D.abcdefgh

7. Which of the declaration is correct?

A.int length;

B.char int;

C.int long;

D.float double;

8. Which of the following statements are correct about the below C-program?

```
#include<stdio.h>
int main()
{
    int x = 10, y = 100%90, i;
    for(i=1; i<10; i++)
    if(x != y);
        printf("x = %d y = %d\n", x, y);
    return 0;
}
```

1: The printf() function is called 10 times.

2: The program will produce the output x = 10 y = 10

3: The ; after the if(x!=y) will NOT produce an error.

4: The program will not produce output.

A. 1

B. 2, 3

C. 3, 4

D. 4

9. What will be the output of the program?

```
#include<stdio.h>
int main()
{
    int k, num=30;
    k = (num>5 ? (num <=10 ? 100 : 200) : 500);
    printf("%d\n", num);
    return 0;
}
```

A. 200

B. 30

C. 100

D. 500

10. If **char=1**, **int=4**, and **float=4** bytes size, what will be the output of the program?

```
#include<stdio.h>

int main()
{
    char ch = 'A';
    printf("%d, %d, %d", sizeof(ch), sizeof('A'), sizeof(3.14f));
    return 0;
}
```

A. 1, 2, 4

B. 1, 1, 4

C. 2, 2, 4

D. 2, 4, 8

11. What will be the output of the program?

```
#include<stdio.h>

int main()
{
    char str[] = "Nagpur";
    str[0]='K';
    printf("%s, ", str);
    str = "Kanpur";
    printf("%s", str+1);
    return 0;
}
```

A. Kagnpur, Kanpur

B. Nagpur, Kanpur

C. Kagnpur, anpur

D. The program has an error

12. Which of the following statement is correct about the program?

```
#include<stdio.h>

int main()
{
    FILE *fp;
    char ch;
    int i=1;
    fp = fopen("myfile.c", "r");
    while((ch=getc(fp))!=EOF)
    {
        if(ch == '\n')
            i++;
    }
    fclose(fp);
    return 0;
}
```

A. The code counts number of characters in the file

B. The code counts number of words in the file

C. The code counts number of blank lines in the file

D. The code counts number of lines in the file

13. Which of the following statements is correct about the program?

```
// All necessary have been included
int main()
{
    FILE *fp = NULL;
    char name[100] = "Le Thi Ca Chua";
    int MAX = 100;
    int r;
    fp = fopen("myfile.txt", "r+");
    if (fp != NULL) {
        fgets(name, MAX, fp);
        fprintf(fp, "%s", "New Data");
        fclose(fp);
    } else {
        printf("Failed to open file\n");
    }

    getch();
    return 0;
}
```

A. The program will read data and then append "New Data" to the file myfile.txt

B. The program will read data and then override old data with "New Data" in the file myfile.txt

C. The program will erase all data in the file myfile.txt

D. The program can only read data from the file myfile.txt

14. Which symbol is used to make comments?

A. //

B. #

C. !!

D. <!--

15. Which of the following statements is used to insert pre-written code into a current program?

A. #read <file>

B. #get <file>

C. #include <file>

D. #pre <file>

16. Which symbols represent a block of code ?

A. { ... code here }

B. (... code here)

C. [... code here]

D. < ... code here >

17. In which standard library file is the function printf() located?

A. stdlib.h

B. stdio.h

C. stdout.h

D. stdoutput.h

18. Which mathematical operators are in the correct order?

A. Addition, Division, Modulus

B. Division, Multiplication, Modulus

C. Modulus, Multiplication, Subtraction

D. Modulus, Addition, Division

19. Select a correct statement:

A. Variables created in a function cannot be used another function.

B. Variables created in a function can be used in another function.

C. Variables created in a function can only be used in the main function.

D. None of the others.

20. Which symbol is used to declare a pointer?

A. &

B. @

C. **

D. \$

21. Which symbol is used to reference a pointer?

A. *

B. &

C. @

D. \$

22. if we use two's complement technique, what is the binary form of -92?

A. 10100100

B. 01011001

C. 10110100

D. 10111111

23. Which of the declaration is correct?

A. int 3length;

B. char c = "char";

C. int lo-ng;

D. None of the others.

24. Which of the following options of the FILE *fopen(char file_name[], char mode[]) allows user to open and append data to a file?

A. "a"

B. "append"

C. "r"

D. "w"

25. Which of the following functions moves the file pointer to the beginning of the file?

A. rewind()

B. begin()

C. moveToBegin()

D. start()

26. Select a correct statement about the program below:

```
#include<stdio.h>
int main()
{
    char ch;
    while(x=0;x<=255;x++)
        printf("ASCII value of %d character %c\n", x, x);
    return 0;
}
```

A. The code generates an infinite loop

B. The code prints all ASCII values and its characters

C. while() statement syntax error

27. What is the output of the program below?

```
#include<stdio.h>
#include<math.h>
int main()
{
    float n=1.54;
    printf("%f, %f\n", ceil(n), floor(n));
    return 0;
}
```

A. 2.000000, 1.000000

B. 1.5000, 1.500000

C. 1.550, 2.0000

D. 1.000000, 2.0000

28. What will be the output of the program?

```
#include<stdio.h>

int main()
{
    float arr[] = {12.4, 2.3, 4.5, 6.7};
    printf("%d\n", sizeof(arr)/sizeof(arr[0]));
    return 0;
}
```

A. 5

B. 4

C. 6

D. 1

29. A string is:

A. an array of characters that has a null-byte character after its last meaningful element.

B. an array pointer of characters that has a null-byte character after its last meaningful element.

C. an array of strings that has a null-byte character after its last meaningful element.

D. an array pointer of characters.

30. Which of the following testing techniques that we don't need to care about the knowledge of the internal structure?

A. White box

B. Black box

C. Branch testing

D. Data flow testing

31. What does the following program do:

<pre>#include <stdio.h> char *program = "#include <stdio.h>%cchar *program = %c%c%c;%cint main()%c{%c printf(program, 10, 34, program, 34, 10, 10, 10, 10, 10, 10);%c int main(int argc, char *argv[]) { printf(program, 10, 34, program, 34, 10, 10, 10, 10, 10, 10); system("PAUSE"); return 0; }</pre>	<pre>return 0;%c}%c";</pre>
---	-----------------------------

A. The program will print out nothing

B. The program will print the source code itself

C. I don't know. What the hell is this?

D. Compile error

32. What should be placed at line 4?

```
2 #include <stdio.h>
3 #include <conio.h>
4 // Missing line
5
6 int main()
7 {
8     char ar_str[80];
9     int i = 0, count = 0;
10
11     printf("\nEnter a string : ");
12     gets(ar_str);
13
14     while (ar_str[i] != 0)
15     {
16         if (isalpha(ar_str[i++]))
17             count++;
18     }
19
20     printf("%d\n", count);
21     getch();
22     return 0;
23 }
```

A. #include <ctype.h>

B. Do not need to place anything. The program is perfectly correct.

C. #include <conio.h>

D. #include <stdlib.h>

Q33. What is the output?

```

#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
#include <string.h>
char *f(char *s)
{
    char *tmp, i;

    i = 0;
    tmp = (char *)malloc(strlen(s)+1);
    while (i<strlen(s))
        *(tmp+i) = *(s + strlen(s) - i++ - 1);
    *(tmp+i) = 0;
    return tmp;
}

int main()
{
    char hello[] = "Hello World";
    char *s;

    s = f(hello);
    printf("\n%s", s);
    getch();

    return 0;
}

```

A. dlroW olleH

B. dlroW

C. Hello

D. World

Q34. What is the output?

```

#include <stdio.h>
#include <conio.h>

unsigned long f(int n)
{
    unsigned long r = 1;
    int i;
    for (i=2; i<=n; i++)
    {
        r *= i;
    }
    return r;
}

int main()
{
    int n = 5;

    printf("\n%lu", f(n));
    getch();

    return 0;
}

```

A. 12

B. 120

C. Compile error

D. Runtime error

Q35. What is the output?

```

#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    int i;
    for(i=0;i<5;i++){
        int i=10;
        printf(" %d",i);
        i++;
    }
    system("PAUSE");
    return 0;
}

```

A. 10 10 10 10 10

B. 10 11 12 13 14

C. 0 1 2 3 4

D. Compile error

Q36. What is the output?

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
    for(;;NULL;)
    {
        printf("cquestionbank");
    }
    system("PAUSE");
    return 0;
}
```

A. cquestionbank

B. Nothing will be printed out.

C. Infinite loop

D. Compile error

Q37. What are the outputs?

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[])
{

    int i = 0;
    int n = 40;
    int *p = &i;
    i = 20;

    if(*p == 10)
    {
        n = n/*p;
    }

    printf("%d", n);

    system("PAUSE");
    return 0;
}
```

A. 4

B. 2

C. 1

D. Compile error

Q38. What is the output?

```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[])
{

    int i = 0;
    int n = 40;
    int *p = &i;
    int **pp = &p;
    i = 20;
    *pp++;
    *p++;

    printf("%d", *p);

    system("PAUSE");
    return 0;
}
```

A. 21

B. 41

C. An Unpredictable result. It could be a garbage value.

D. Compile error

Q39. Which of the following code prints out the current time of the system?

A.

```
time_t lt;
```

```
lt = time(NULL);
```

```
printf("%s\n", ctime(&lt));
```

B.

```
time_t lt;
```

```
lt = time(SYSTEM);
```

```
printf("%s\n", ctime(&lt));
```

C.

```
time_t lt;
```

```
lt = time(TIME);
```

```
printf("%s\n", ctime(&lt));
```

D.

```
time lt;
```

```
lt = time(NULL);
```

```
printf("%s\n", ctime(lt));
```

Q40. Consider the following code:

```
int main(int argc, char *argv[])
```

```
{
```

```
    int x, y;
```

```
    printf("Input 2 integers:");
```

```
    scanf("%d,%d,%d", &x, &y);
```

```
    printf("%d\n", x + y);
```

```
    return 0;
```

```
}
```

Which of the following values should be inputted so that the program will produce the result of 7?

A. 1,2,4

B. 5423,3,4

C. 3,4,0

D. 1 2 4

Q41. In terms of software testing, which of the following topics that quality is constrained to?

A. Correctness

B. Completeness

C. Security

D. All of the others

Q 42. Black box testing is a technique that focuses on:

A. External factors

B. Internal structures

C. Internal code

D. None of the others

Q 43. Which of the following is/are white-box testing forms?

A. Control flow testing

B. Data flow testing

C. Branch testing

D. Stressed testing

Q 44. Which of the following is not a phase in waterfall model?

A. Design

B. Coding

C. Deployment

D. None of the others

Q 45. Which of the following software development model focuses on user-driven rather than concept-driven?

A. Open Source Model

B. WaterFall Model

C. Practical Model

D. None of the others